

TrES-3b

R-1852

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150408 · created 2026-08-03T08:22:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457120.9127 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.016
Transit depth (Rp/Rs)^2	1.81 +/- 0.43 [%]
Orbital Inclination (inc)	83.05 +/- 0.99
Airmass coefficient 1 (a1)	1.158 +/- 0.011
Airmass coefficient 2 (a2)	-0.1233 +/- 0.007
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	None
Optimal Aperture	3.19
Optimal Annulus	11.65
Transit Duration (day)	0.0611 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.135 ± 0.016 vs 0.16309 (deviation 1.55σ)
Duration fit vs published	0.0611 d vs 0.0567 d (deviation 0.53σ)
Mid-time O–C	0.7 min
Depth SNR	7.4
Residual scatter	0.92 %

Light curve

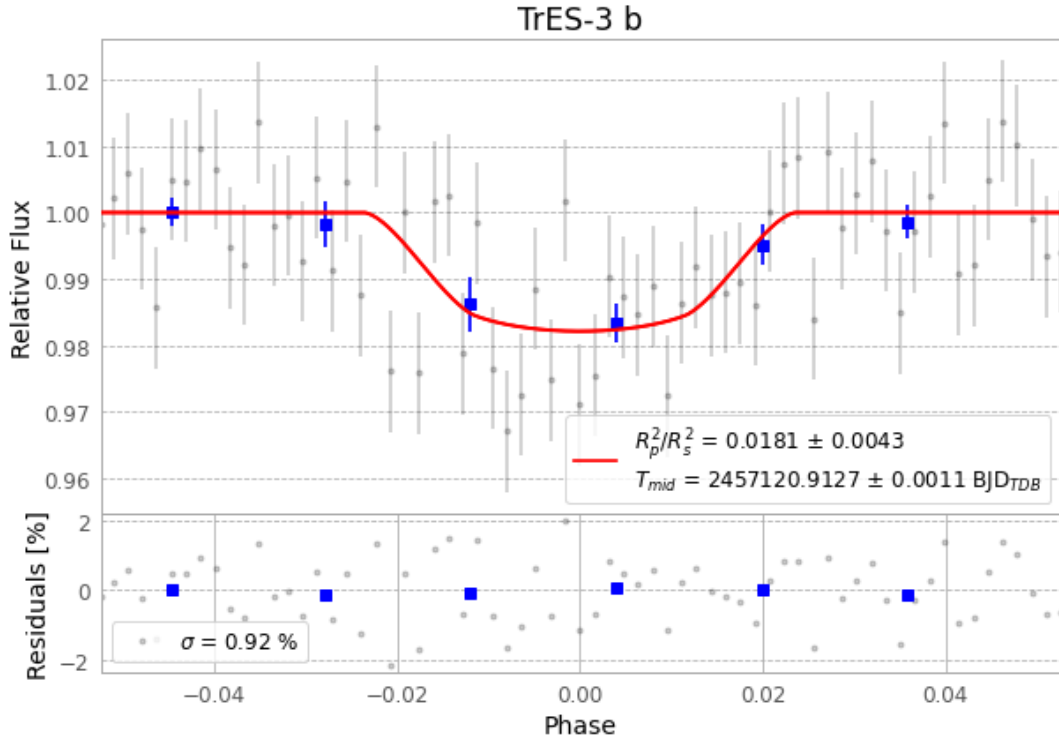


Figure 1 — FinalLightCurve_TrES-3 b_2015-04-08.png (SHA-256 aae90eb23048c749...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [245, 259], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 74373de239fed6b2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), TRES-3150408081243.FITS → TRES-3150408113012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150408.log (10552028714c...), FinalLightCurve_TrES-3 b_2015-04-08.png (aae90eb23048...), FinalLightCurve_TrES-3 b_2015-04-08.csv (81b5261dea76...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 08:22Z.